DAY 35 | POST-READ

Drift and Optimisation: Recap and Beyond

What we covered today, plus ideas to carry into the capstone.

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Part A: Summary of today

Topic	What to remember
Drift types	Data (inputs), concept (correct answers), knowledge (documents), model (behaviour), output.
PSI	Compare bucket shares between periods; under 0.1 stable, over 0.25 significant (rules of thumb).
KS test	Compares two numeric distributions; small p-value means they differ.
Embedding drift	Distance between average question vectors per period; cluster to see what is new.
Concept drift fix	Update golden answers first, then documents and prompts.
Optimisation	Hypothesis, single change, measure on golden set, decide, canary release.
Levers	Retrieval, prompt, model routing, data. Data fixes often win.
Automation	Ingestion, drift reports, evaluation, model version checks, stakeholder reports on a schedule.

Part B: Going beyond today

1. Statistical significance with big data

With millions of requests, tiny meaningless differences become "significant". Look at effect size (how big the change is) as well as p-values, and confirm with examples.

2. Topic discovery

Cluster recent questions (for example k-means on embeddings) and ask a model to name each cluster. New clusters with high refusal rates show exactly which documents are missing.

3. Automated prompt optimisation

Tools can search for better prompts against a golden set. Use them carefully: optimising too hard on a small set overfits. Keep a held-out set the optimiser never sees.

4. Scheduling

Use cron, Azure Container Apps jobs, Azure DevOps scheduled pipelines or Airflow to run the weekly job. Store each report so trends are visible over months.

5. Carrying this into the capstone

Your capstone should include at least: a golden set, a quality gate, structured logs, one dashboard and one drift or quality check. Evaluators will look for them.

Part C: Check yourself

- What is the difference between data drift and concept drift?
- How do you read a PSI of 0.3?
- Why update golden answers before documents when a policy changes?
- Why change one variable per experiment?
- Name three jobs worth scheduling weekly.

Part D: Practice ideas for the weekend

- Read all five capstone briefs and rank your top two with reasons.
- List which notebooks from Days 1 to 35 you will reuse for your chosen project.
- Draft your capstone's golden set with 10 items.

Coming up: the capstone (Days 36 to 40). Five real-world project options, built end to end in teams, evaluated with a shared rubric.

Yoi shuumatsu wo (良い週末を): have a great weekend